

HOMEWORK 4

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1. Let X be an infinite set. For $p \in X$ and $q \in X$, define

$$d(p, q) = \begin{cases} 1 & (\text{if } p \neq q) \\ 0 & (\text{if } p = q) \end{cases}$$

Prove that this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?

Proof. If $p = q$, then $d(p, q) = 0$. If $p \neq q$, then $d(p, q) = d(q, p) = 1 > 0$. Let $t \in X$ such that $d(p, q) \leq d(p, t) + d(t, q)$. If $p = q$, then $0 \leq d(p, t) + d(t, q)$, so it holds. If $p \neq q$, then $1 \leq d(p, t) + d(t, q)$. Suppose $d(p, t) + d(t, q)$ is zero. This forces $t = p = q$, which is false, so $d(p, t) + d(t, q) \geq 1$. Thus, this is a metric. Let $A \subseteq X$. To prove openness, $\exists r > 0$ such that $B_r(p) \subseteq A$. Let $r < 1$. Thus, we get $B_r(p) = \{p\} \subseteq A$, so this is true. Since A was arbitrary, every subset of X is open. Since A^c is also an arbitrary subset of X , the same argument shows A^c is open. By definition, a set is closed when its complement is open. So, A is closed. Since A was arbitrary, every subset of X is closed. For compactness, if $K \subseteq X$ is finite, any open cover of K has a finite subcover by picking an open cover for a singular point in K . If K is infinite, consider the cover $\{\{p\} : p \in K\}$; each singleton is open, but since there are infinitely many points, there exists no finite open cover. Therefore, the compact subsets of X are the finite subsets. $\square$

2. Let $A_1, A_2, \dots$ be subsets of a metric space.

- a) If $B_n = \bigcup_{i=1}^n A_i$, prove that $\bar{B}_n = \bigcup_{i=1}^n \bar{A}_i$, for $n = 1, 2, 3, \dots$
- b) If $B = \bigcup_{i=1}^\infty A_i$, prove that $\bigcup_{i=1}^\infty \bar{A}_i \subset \bar{B}$

Show, by an example, that this inclusion can be proper, that is,

$$\bar{B} \supset \bigcup_{i=1}^\infty \bar{A}_i, \quad \bar{B} \neq \bigcup_{i=1}^\infty \bar{A}_i$$

Proof. Use induction. The base case is simple: $B_1 = A_1 \implies \bar{B}_1 = \bar{A}_1$. Now assume $1, \dots, n$ is true, and deduce $n + 1$. Thus, we have $B_{n+1} = \bigcup_{i=1}^{n+1} A_i \implies B_{n+1} = B_n \cup A_{n+1}$. Now we must prove $\bar{B}_{n+1} = \bar{B}_n \cup \bar{A}_{n+1}$.

Lemma 1. $\overline{X \cup Y} = \bar{X} \cup \bar{Y}$

Proof. $\overline{X \cup Y} = (X \cup Y) \cup (X \cup Y)'$ and $\bar{X} \cup \bar{Y} = (X \cup X') \cup (Y \cup Y')$. Unions hold the commutative property, so $(X \cup X') \cup (Y \cup Y') = (X \cup Y) \cup (X' \cup Y')$. Now we show that $X' \cup Y' = (X \cup Y)'$. Let $p \in X'$, thus it is a limit point for X . Since it is a limit point of X , then it is a limit point for $X \cup Y$. Thus, $p \in (X \cup Y)'$. Let $q \in Y'$. Thus, it is a limit point for Y . Since it is a limit point for Y , we may deduce it is a limit point for $X \cup Y$. Therefore, $q \in (X \cup Y)'$. Thus, $X' \cup Y' = (X \cup Y)'$. Conversely, let $p \in (X \cup Y)'$ but $p \notin X'$ and $p \notin Y'$. Since $p \notin X'$, there is a neighborhood N_1 with

$N_1 \cap X \subseteq \{p\}$. Since $p \notin Y'$, there is a neighborhood N_2 of p with $N_2 \cap Y \subseteq \{p\}$. Let $N = N_1 \cap N_2$, taking the smaller radius. Then, $N \cap (X \cup Y) \subseteq \{p\}$, so N has no point of $X \cup Y$ other than p , which contradicts $p \in (X \cup Y)'$. Now, referring to our original equation, $(X \cup Y) \cup (X \cup Y)' = (X \cup X') \cup (Y \cup Y')$. Rearranging, we get $(X \cup Y) \cup (X \cup Y)' = (X \cup Y) \cup (Y' \cup X')$. We can take out the $(X \cup Y)$, and since we have shown equivalency, we may deduce that these are the same. $\square$

Using our lemma, we see that $\bar{B}_{n+1} = \overline{B_n \cup A_{n+1}} \implies \bar{B}_{n+1} = \bar{B}_n \cup \bar{A}_{n+1}$. Thus, this is true by induction for part a. $\square$

Proof. Since $B = \bigcup_{i=1}^{\infty} A_i$, we have $A_i \subseteq B$ for any i . $A = \bigcup_{i=1}^n A_i \subseteq B \implies \bar{A} \subseteq \bar{B}$ from the first proof, it follows that $\bar{A}_i \subseteq \bar{B}$ for every i . Since this holds for each i , taking the union over all i gives $\bigcup_{i=1}^{\infty} \bar{A}_i \subseteq \bar{B}$. $\square$

3. Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but E is not compact. Is E open in $\mathbb{Q}$?

Proof. Let $E = \{p \in \mathbb{Q} : \sqrt{2} < p < \sqrt{3}\} \sqcup \{p \in \mathbb{Q} : -\sqrt{3} < p < -\sqrt{2}\}$. Let $p \in E$. If $p > 0$, then $p \in (\sqrt{2}, \sqrt{3})$. If $p < 0$, then $p \in (-\sqrt{3}, -\sqrt{2})$. In either case, we see $|p| < \sqrt{3}$, therefore it is bounded. To show E is closed, let a_n be a sequence in E such that $2 < a_n^2 < 3$. If a_n converges to $a \in \mathbb{Q}$, then $a \in E$. Thus, as $n \rightarrow \infty$, $a_n \cdot a_n \rightarrow a \cdot a$. Thus, since $a_n \in E$ for all n , we are given $2 \leq a^2 \leq 3$. But since $a \in \mathbb{Q}$, we are forced with $2 < a^2 < 3$. Thus, E is closed since all convergent sequences in E converge in E . To show that this is not compact, consider $G_n = \{(-\sqrt{3} - \frac{1}{n}, \sqrt{3} - \frac{1}{n} : n \in \mathbb{N})\}$. We see that if $p > 0$, then it is covered by the positive side of G_n . If $p < 0$, then it is covered by the negative side of G_n . Thus, $E \subseteq \bigcup_{n=1}^{\infty} G_n$. Let's suppose, for contradiction, we only need finitely many G_n such that $E \subseteq \bigcup_{n=1}^N G_n$. If G_n were finite, then we show that in the interval $(\sqrt{3} - \frac{1}{N}, \sqrt{3})$, where N is the max index, there lies a rational p in the interval through the Archimedean property. This is contradictory, since p is rational and is between the interval, that shows $p \in E$, but this element is not covered in the finite open cover. If N is small, we may introduce a new interval $(\max(\sqrt{3} - \frac{1}{N}, \sqrt{2}), \sqrt{3})$, which guards against this. Thus, E is not compact. To show E is open, we must show $E \subseteq E^\circ$. Thus, we must show that for any $p \in E$, $\exists r > 0$ such that $B_r(p) \subseteq E$. But we can immediately deduce that this is true, because the two sets that make up E are open intervals. Therefore, E is open. $\square$

4. Construct and prove a compact set of real numbers whose limits points form a countable set.

Proof. Consider $E = \{\frac{1}{n} : n \in \mathbb{N}\} \cup \{0\}$. Let $p \in E$. Thus, any arbitrary element p satisfies $0 \leq p \leq 1$, so $E \subseteq [0, 1]$. We show $E' = \{0\}$. As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$, so $\{0\}$ is the only limit point of E . Every other point $\frac{1}{n}$ is isolated. We may consider an arbitrary $N > 2 \in \mathbb{N}$ such that $\frac{1}{N-1} > \frac{1}{N} > \frac{1}{N+1}$. We may take $r = \min(\frac{\frac{1}{N} + \frac{1}{N-1}}{2}, \frac{\frac{1}{N} + \frac{1}{N+1}}{2})$, and we are able to construct a neighborhood $N_r(\frac{1}{N})$ that only contains $\frac{1}{N}$. Thus, each point, other than zero, is isolated. Since $E' = \{0\}$, $E' \subseteq E$, therefore E is closed. Since E is closed and bounded, E is compact by theorem 2.41. $\square$